

Q1: [20] Three identical charges of charge Q are located at the vertices of an equilateral triangle of side length a . Determine the force on one of the charges due to the other two.

Solution To determine the force on one of the charges Q located at a vertex of an equilateral triangle due to the other two charges, we can use Coulomb's law. Let the three charges Q be located at the vertices of an equilateral triangle with side length a . Label the charges as Q_1 , Q_2 , and Q_3 , and consider the force on charge Q_1 due to charges Q_2 and Q_3 . Since the triangle is equilateral, each angle between two sides is 60° . By Coulomb's law, the magnitude of the force between any two charges Q separated by a distance a is:

$$F = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{a^2}.$$

The force on Q_1 due to Q_2 , denoted as $\mathbf{F}_{12}$, is repulsive (since both charges are positive) and directed along the line joining Q_1 and Q_2 . This force has magnitude $F = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{a^2}$. Similarly, the force on Q_1 due to Q_3 , denoted as $\mathbf{F}_{13}$, is repulsive and directed along the line joining Q_1 and Q_3 . This force also has magnitude $F = \frac{1}{4\pi\epsilon_0} \frac{Q^2}{a^2}$. Since Q_1 , Q_2 , and Q_3 form an equilateral triangle, the angle between $\mathbf{F}_{12}$ and $\mathbf{F}_{13}$ is 60° . The magnitude of the force on one of the charges due to the other two is:

$$F_{\text{resultant}} = \frac{\sqrt{3}}{4\pi\epsilon_0} \frac{Q^2}{a^2}.$$

The proof is straightforward and relies on basic geometric manipulations. Try working through it on your own.

Q2: [40] A total charge of Q_1 is distributed uniformly along a semicircular ring as shown in Figure 1. Two point charges, each of magnitude Q_2 , are situated as shown. The surrounding medium is free space.

- Find Q_2 in terms of Q_1 so that the potential V at the center of the ring is zero.
- Find Q_2 in terms of Q_1 so that the electric field $\mathbf{E}$ at the center of the ring is zero.

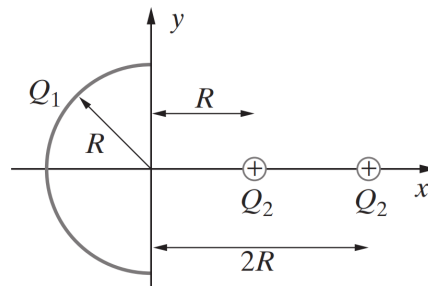


Figure 1: Figure for Question 2.

Solution If ρ_ℓ is the line charge density, the total charge on it is

$$Q_1 = \rho_\ell \times \text{arc_length} = \rho_\ell \times \pi R$$

$$\Rightarrow \rho_\ell = \frac{Q_1}{\pi R}$$

The electrostatic potential at the center due to the ring is

$$V_{\text{ring}}(\text{origin}) = V_{\text{ring}}(0) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho_\ell d\ell}{R} = \frac{1}{4\pi\epsilon_0} \int \frac{Q_1}{\pi R} \frac{R d\phi}{R} = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{\pi R} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\phi = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{R}$$

The potentials at origin due to the charges Q_2 , at R and $2R$ distance from the origin, are

$$V_{P_1}(0) = \frac{1}{4\pi\epsilon_0} \frac{Q_2}{R} \quad \text{and} \quad V_{P_2}(0) = \frac{1}{4\pi\epsilon_0} \frac{Q_2}{2R}$$

Adding the three potential contributions, we attain the net potential at the center of the ring,

$$\begin{aligned} V_{\text{total}}(0) &= \frac{1}{4\pi\epsilon_0} \frac{Q_1}{R} + \frac{1}{4\pi\epsilon_0} \frac{Q_2}{R} + \frac{1}{4\pi\epsilon_0} \frac{Q_2}{2R} \\ &= \frac{1}{4\pi\epsilon_0} \frac{1}{R} \left[Q_1 + \frac{3}{2} Q_2 \right] \end{aligned}$$

If the potential at $(0,0)$ is zero, then it is immediate that $Q_2 = -\frac{2}{3}Q_1$.

The electric field at the origin due to the semi-circle is

$$\mathbf{E}_{\text{ring}} = \frac{1}{4\pi\epsilon_0} \int \frac{Q_1}{\pi R} R d\phi \frac{\mathbf{R}}{R^3}$$

where $\mathbf{R} = \mathbf{r} - \mathbf{r}' = 0 - (R\mathbf{a}_\rho) = -(R \cos \phi \mathbf{a}_x + R \sin \phi \mathbf{a}_y)$

$$\mathbf{E}_{\text{ring}} = \frac{1}{4\pi\epsilon_0} \frac{Q_1}{\pi R^2} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (\cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y) d\phi = \frac{1}{4\pi\epsilon_0} \frac{2Q_1}{\pi R^2} \mathbf{a}_x \quad (\text{do it yourself})$$

The field at origin due to the charges Q_2 , at R and $2R$ distance from the origin, are

$$\mathbf{E}_{P_1}(0) = -\frac{1}{4\pi\epsilon_0} \frac{Q_2}{R^2} \mathbf{a}_x \quad \text{and} \quad \mathbf{E}_{P_2}(0) = -\frac{1}{4\pi\epsilon_0} \frac{Q_2}{4R^2} \mathbf{a}_x$$

Gathering these fields together and summing them up we obtain the net field at point $P(0,0)$, we get

$$\begin{aligned} \mathbf{E}(\text{origin}) &= \frac{1}{4\pi\epsilon_0} \frac{2Q_1}{\pi R^2} \mathbf{a}_x - \frac{1}{4\pi\epsilon_0} \frac{Q_2}{R^2} \mathbf{a}_x - \frac{1}{4\pi\epsilon_0} \frac{Q_2}{4R^2} \mathbf{a}_x \\ &= \frac{1}{4\pi\epsilon_0} \frac{1}{R^2} \left(\frac{2Q_1}{\pi} - \frac{5}{4} Q_2 \right) \mathbf{a}_x \end{aligned}$$

If $\mathbf{E}(\text{origin})$ is null, then we can express Q_2 in terms of Q_1 , as follows:

$$Q_2 = \frac{8Q_1}{5\pi}.$$

Q3: [40] The space between two concentric spheres of radii a and b ($a < b$) in free space is charged to a volume charge density given by

$$\rho(r) = \frac{K}{r^2}; \quad a \leq r \leq b$$

where K is a constant. (a) Find the total charge in the shell. (b) Find the electric field at all points in space. (c) Find the electric potential at all points in space. (d) What happens if $b \rightarrow a$?

Solution: (a) To calculate the total charge which is within the shell, we apply the integral

$$Q_{\text{total}} = \int_V \rho_v dv = \int_a^b \frac{K}{r^2} 4\pi r^2 dr = 4\pi K(b - a), \quad a < b.$$

(b) For $0 < r \leq a$, $\mathbf{E} = 0$ because the internal contributions of $\mathbf{E}$ from $r = a$ towards the center cancel each other, which may easily noticed by applying Gauss law in this region. For $a \leq r \leq b$, by applying Gauss law,

$$\begin{aligned} \text{LHS} &= \frac{Q_{\text{enc}}}{\epsilon_0} = \int_V \rho_v dv = \int_a^r \frac{K}{r'^2} 4\pi r'^2 dr' = 4\pi K(r - a), \quad a \leq r \leq b \\ \text{RHS} &= \oint \mathbf{E} \cdot d\mathbf{S} = 4\pi r^2 E_r \\ \implies E_r &= \frac{Q_{\text{enc}}}{4\pi\epsilon_0 r^2} = \frac{K(r - a)}{\epsilon_0 r^2}, \quad a \leq r \leq b \end{aligned}$$

For $r \geq b$, we take a spherical Gaussian surface, concentric with the spheres of the shell, but with radius $r \geq b$. Outside the shell, the charge enclosed by the Gaussian surface is the total charge Q_{total} . Hence, by applying Gauss's law,

$$\begin{aligned} \frac{Q_{\text{total}}}{\epsilon_0} &= \oint \mathbf{E} \cdot d\mathbf{S} = 4\pi r^2 E_r, \\ \implies E_r &= \frac{Q_{\text{total}}}{4\pi\epsilon_0 r^2} = \frac{K(b - a)}{\epsilon_0 r^2}, \quad r \geq b \end{aligned}$$

To determine $V(r)$ for $r \geq b$ The relation between the electrostatic field and the electric potential for this case is

$$V(r \geq b) = - \int_{\infty}^r E_r(r' \geq b) dr' = - \int_{\infty}^r \frac{K(b - a)}{\epsilon_0 r'^2} dr' = \frac{K(b - a)}{\epsilon_0 r}, \quad r \geq b$$

When $r = b$, then we have $V(b) = \frac{K(b - a)}{\epsilon_0 b}$.

To determine $V(r)$ for $a \leq r \leq b$, we have

$$\begin{aligned} V(r) &= - \int_b^{r(a \leq r < b)} E_r(a \leq r' \leq b) dr' - \int_{\infty}^b E_r(r' \geq b) dr' \\ &= \int_r^b \frac{K(r' - a)}{\epsilon_0 r'^2} dr' + V(b) \\ &= \frac{K}{\epsilon_0} \left[a \left(\frac{1}{b} - \frac{1}{r} \right) + \ln(b) - \ln(r) \right] + V(b) \\ &= \frac{K}{\epsilon_0} \left[a \left(\frac{1}{b} - \frac{1}{r} \right) + \ln(b) - \ln(r) \right] + \frac{K(b - a)}{\epsilon_0 b}. \end{aligned}$$

Similarly, one may easily obtain the following for $r < a$:

$$V(r \leq a) = \frac{K(b - a)}{\epsilon_0 b} + \frac{K}{\epsilon_0} \left[a \left(\frac{1}{b} - \frac{1}{a} \right) + \ln \left(\frac{b}{a} \right) \right] = \text{constant}.$$